The period during which a return is earned or computed can vary and often we have to annualize a return that was calculated for a period that is shorter (or longer) than one year. You might buy a short-term treasury bill with a maturity of three months, or you might take a position in a futures contract that expires at the end of the next quarter. How can we compare these returns?

In many cases, it is most convenient to annualize all available returns to facilitate comparison. Thus, daily, weekly, monthly, and quarterly returns are converted to annualized returns. Many formulas used for calculating certain values or prices also require all returns and periods to be expressed as annualized rates of return. For example, the most common version of the Black–Scholes option-pricing model requires annualized returns and periods to be in years.

Non-annual Compounding

Recall that interest may be paid semiannually, quarterly, monthly, or even daily. To handle interest payments made more than once a year, we can modify the present value formula as follows. Here, R_s is the quoted interest rate and equals the periodic interest rate multiplied by the number of compounding periods in each year. In general, with more than one compounding period in a year, we can express the formula for present value as follows:

$$PV = FV_N\left(1 + \frac{R_s}{m}\right)^{-mN}, \tag{7}$$

where

m = number of compounding periods per year,

R_s = quoted annual interest rate, and

N = number of years.

The formula in Equation 8 is quite similar to the simple present value formula. As we have already noted, present value and future value factors are reciprocals. Changing the frequency of compounding does not alter this result. The only difference is the use of the periodic interest rate and the corresponding number of compounding periods.

The following example presents an application of monthly compounding.

EXAMPLE 11

The Present Value of a Lump Sum with Monthly Compounding

The manager of a Canadian pension fund knows that the fund must make a lump-sum payment of CAD5 million 10 years from today. She wants to invest an amount today in a guaranteed investment contract (GIC) so that it will grow to the required amount. The current interest rate on GICs is 6 percent a year, compounded monthly.

1. How much should she invest today in the GIC?

   Solution:

   By applying Equation 8, the required present value is calculated as follow: